INDIE DEVELOPER TRACK

CAIRN

Object permanence for physical work.

> The first AI that can take something apart and **put it back together**.

SAM 3.1 · Muse Wonder · Muse Glimmer · Muse Voice · Muse Image



REPAIR PASSPORT

Fasteners accounted	**23 / 23**
Extra parts	**0**
Boot test	**passed**

SHA-256 · signed 14:32 · bench 02

THE THESIS

AI has object recognition. It does not have object permanence. **Cairn gives it a persistent memory of every part on the bench** — and the proof that the job was done right.

PART I · PP. 02–08
The case
The game, the problem, the idea, the rubric, the landscape, the demo.

PART II · PP. 09–15
The build
Architecture, object graph, stack, ten-day plan, 48-hour cut, eval, risk.

PART III · PP. 16–18
The win
Business, words, and the receipts — everything the submission needs.

THEME	BUILD WINDOW	PRIZE POOL	PREPARED
Build for Your World Push the Frontier	26 Oct – 4 Nov 2026 10 days · virtual	$1,000,000 Indie + Startup	2 November 2026 Blueprint v1.0

The missing sense is memory.

THESIS
SEC. 01

AI can recognize a screw. It cannot remember that the screw exists once it leaves the frame. That is why AI can design a phone but cannot repair one.

01

The problem

Repair is a memory problem. A technician builds a fragile mental model of a device, then loses it when they leave the bench. Parts go missing, steps get skipped, and there is no proof the job was done right.

02

The insight

Models have object recognition but not object permanence. Every physical task is a state machine over objects. Current models have no state. Give them one and a new class of agent becomes possible.

03

The product

Cairn watches a repair bench, tracks every part with identity, remembers the teardown across time, catches missing or wrong parts, and certifies reassembly with a signed repair passport.

> "Object recognition is solved. Object permanence is not. Cairn is the first AI that can take something apart and put it back together."

THE ONE-SENTENCE PITCH

62M t
E-WASTE GENERATED / YEAR

22.3%
FORMALLY RECYCLED

30%
IMPACT WEIGHTING

25%
DEMO WEIGHTING

The 30-second pitch

"Today's AI can tell you what a screw is. It can't tell you which hole it came from. Cairn gives AI object permanence: a live, persistent memory of every part on a repair bench. It uses SAM 3.1 to track parts, Muse Wonder to reason over the object graph, and Muse Glimmer to run locally so customer video never leaves the shop. Disassemble a laptop, drop a screw, skip the battery step, walk away — Cairn remembers everything and proves the reassembly with a signed repair passport."

Why this wins

- **Impact:** repair, right-to-repair, e-waste, and the skilled-trades gap — a universal problem with a specific beneficiary.
- **Meta tooling:** SAM 3.1 is the hero; Wonder, Glimmer, Voice, and Image each have a necessary role.
- **Depth:** persistent graph, re-ID, safety kernel, verification invariants, eval harness.
- **Demo:** physical, visual, interactive, and robust — the judges will remember it.

How to read this blueprint

Pages 03–08 are the case: the game, the problem, the idea, the rubric, the landscape, and the demo. **Pages 09–15** are the build: architecture, data model, stack, plan, 48-hour cut, eval, and risk. **Pages 16–18** are the win: business, words, and receipts.

The blueprint is written to be executed by one strong builder, or a team of four, without changing the core thesis.

THE BET

The next wave of AI is not more intelligence. It is memory, verification, and agency in the physical world.

Memory
A persistent object graph that survives occlusion, time, and a lunch break.

Verification
Every claim has an evidence frame; every repair ends with a signed passport.

Agency
The agent plans, asks, observes, and recovers — but the human stays the actuator.

It tracks

6–8 part classes with stable identities across occlusion and a 30-minute teardown.

It remembers

State survives a lunch break. Ask "what is missing?" and the graph answers from evidence.

It proves

A signed passport that a shop, an insurer, or a buyer can verify without trusting the technician.

What actually wins.

This is not a “build an AI app” hackathon. It is a “show a new capability” hackathon. The judges are Meta engineers who have seen five hundred chatbots.

SIGNAL	WHAT IT MEANS
Meta Global AI Developer Hackathon	10 days, $1M, virtual, two tracks: Indie (individuals/students) and Startup (teams/founders).
Two themes	Build for Your World (bring models into a domain you know) and Push the Frontier (software, robotics, manufacturing, energy, materials).
Rubric	Impact **30%**, Use of Meta AI tooling **25%**, Technical depth **20%**, Demo **25%**.
Explicitly banned	Medical/mental-health advice, basic RAG, simple image analyzers, Streamlit apps, dashboards, education chatbots, job screeners, nutrition coaches, personality analyzers, sports coaches.
Bonus pools	$20k Most Viral, $10k Best Open-Weights Build, $10k Community Vote, $5k Best Agentic Use-Case, $5k Best Creative Build.

The scoring bars

Impact — real-world potential, who benefits 30%

Tooling — beyond a basic integration 25%

Demo — works live, video worth watching 25%

Depth — sound, refined engineering 20%

The anti-project list

If your idea fits one of these chips, it needs a meaningfully different problem, user, and execution — or it will be discounted before the demo.

- Medical advice
- Mental-health advisor
- Skin-blemish detector
- Basic RAG
- Simple image analyzer
- Streamlit app
- Dashboard-first
- Education chatbot
- Job screener
- Nutrition coach
- Personality analyzer
- Sports coach

The meta

The winners will not be the teams with the most features. They will be the teams that make the judges feel something the models could not do before — and then prove it in under three minutes. **Demo is 25%, but it is also the gate to Impact.** If the judges cannot see it, they cannot score it. Build for the video.

Don’t compete on intelligence

Everyone can call a model. Compete on **state**: what the system remembers, verifies, and can prove.

Don’t build a platform

Build one wedge with a platform story. Ten days rewards focus, not surface area.

Don’t demo a chat window

Put a physical object on a bench. The judges will remember the object, not the prompt.

The path to the prize

Apply / accept → Build 10 days → Submit Nov 4 → Judge Nov 4–8 → Finals Nov 9

Repair is a memory problem.

A technician disassembles a device and creates a temporary, fragile model of it in their head. The model evaporates when they leave the bench — and with it, the confidence that the job was done right.

01

Lost parts

A dropped screw is a silent failure. The device may still boot, but the next owner inherits a hidden defect.

02

Wrong order

Skipping the battery disconnect can short a board. Order matters, and memory is the only thing enforcing it.

03

No proof

"I fixed it" is not evidence. Shops, insurers, and refurbishers need a record of what was done and what was checked.

04

No transfer

Senior techs carry tacit knowledge. When they leave, it leaves. New techs learn by trial, error, and damaged devices.

The scale

62M t	E-waste generated in 2022 — the fastest-growing waste stream in the world.
22.3%	Formally collected and recycled. The rest is landfill, informal recycling, or hoarded.
10+	US states with right-to-repair laws, plus EU ecodesign rules. Access is expanding; capability is not.
1.9M	Projected unfilled US manufacturing jobs by 2033 — the same skilled-trades squeeze hitting repair.

Who feels it most

Independent repair shops. Thin margins, high turnover, training takes 6–12 months. A single lost screw or wrong-order mistake costs a part, a customer, or a board.

ITAD and refurbishers. They process thousands of devices and need per-device proof for warranty, resale grade, and compliance. Manual inspection does not scale.

DIYers and right-to-repair advocates. They have the manual, but not the memory. They need a second pair of eyes that knows what they just did.

The before / after

Before: "I think I have all the screws. I think I disconnected the battery. I think it's fine."

After: "23 of 23 fasteners accounted for. Battery disconnect logged at 14:02. No extra parts. Boot test passed. Passport signed."

Why now

Perception became a commodity

SAM 3.1 does zero-shot, text-prompted detection and identity-preserving tracking. No training data per device. The eye is finally cheap.

Agents became reliable

Muse Wonder plans over long horizons, calls tools, and recovers from errors. A 20-minute repair loop is now feasible.

Local became viable

Muse Glimmer 30B runs on a single GPU. Private, offline, always-on. The shop can keep customer video on the bench.

> "Every repair is a small act of memory. We built the tools, the manuals, and the training — never the memory."

THE GAP CAIRN FILLS

What the technician actually needs

- **Recall:** what came out, from where, and in what order.
- **Order:** the one step that must happen before the next.
- **Proof:** something to show the customer, the insurer, or the next tech.

Give the model a new sense.

Object recognition

"That is a screw." The model labels a region in a frame. The next frame is a new world. Nothing persists.

frame → label → forget

Object permanence

"That is screw #04, removed at 14:02, now in tray slot 4, belongs to the hinge bracket, and is still missing from the reassembly."

frame → identity → state → memory

Cairn is a live repair copilot built on a persistent object graph. It does not just see the bench; it remembers it, reasons over it, and verifies the outcome.

01

See

SAM 3.1 detects, segments, and tracks every part from short text prompts — zero-shot, no training data.

02

Remember

A persistent graph stores part identity, location, state, relationships, and the evidence frame for every claim.

03

Reason

Muse Wonder plans the teardown, answers questions, adapts to interruptions, and explains why a step matters.

04

Talk

Muse Voice Transcribe enables hands-free, push-to-talk interaction. No keyboard with greasy hands.

05

Verify

Invariants + a hard-coded safety kernel check the reassembly and produce a signed repair passport.

06

Stay local

Muse Glimmer runs the always-on watcher on-device, so customer video never leaves the shop.

The demo promise

Disassemble a laptop. Drop a screw. Skip the battery step. Walk away for lunch. **Cairn remembers everything — and proves the reassembly.**

The category

Object permanence for physical work. Repair is the wedge; assembly, maintenance, lab protocols, surgery, and manufacturing are the platform.

The tagline

"The first AI that can take something apart and put it back together." Concrete, memorable, and provable in one demo.

What it sounds like

You

"What's next?"

Cairn

"Disconnect the battery. Then the SSD. The battery is still connected."

You

"I dropped a screw."

Cairn

"That's the hinge screw. Check the floor mat; tray 4 is empty."

You

"Am I done?"

Cairn

"23 of 23 fasteners. No extra parts. Run the boot test."

Scored against the rubric, not the hype.

RUBRIC MAP
SEC. 05

Impact — 30%

- 62M tonnes of e-waste a year; only ~22% formally recycled.
- Right-to-repair laws are expanding; repair capability is not.
- Independent shops and ITAD firms cannot train techs fast enough.
- Specific beneficiary: the technician, the shop, and the person who cannot afford a replacement.

Use of Meta tooling — 25%

- **SAM 3.1** is the hero: zero-shot text-prompt tracking with identity preservation.
- **Muse Wonder** does long-horizon planning and tool use over the graph.
- **Glimmer** makes it local and private; **Voice** makes it hands-free; **Image** makes the passport legible.

Technical depth — 20%

- Persistent object graph, re-ID across occlusion, evidence per claim.
- Deterministic task state machine with LLM "muscles."
- Hard-coded safety kernel the model cannot override.
- Verification invariants + a real eval harness with 10 recorded sessions.

Demo — 25%

- Physical object, live overlay, real consequences.
- Three hero moments: the return, the wrong screw, the passport.
- Chaos test: drop, skip, ask, walk away.
- Designed for a recorded 3-minute video that holds up on replay.

The bonus sweep: one project, five pools

BONUS	VALUE	HOW CAIRN QUALIFIES
Most Viral	$20,000	"AI remembers every screw" is a 30-second physical demo with a universal hook.
Best Open-Weights Build	$10,000	Muse Glimmer 30B running locally/on Modal as the private, always-on bench watcher.
Community Vote	$10,000	Repair is universally relatable: everyone has a broken device and a lost screw.
Best Agentic Use-Case	$5,000	A genuine long-horizon loop: plan, call tools, observe, recover, verify, issue an artifact.
Best Creative Build	$5,000	Muse Image generates the annotated repair passport and step diagrams.

The strategic read

Cairn is designed to be **eligible for five bonus pools without diluting the main track**. The core demo is the repair loop; the local deployment, the agentic long-horizon run, the creative passport, and the viral hook are all natural extensions of the same build.

The four questions the judges will ask

"Is this just SAM + a UI?"

No. SAM is the eye. The product is the persistent graph, the safety kernel, and the passport. Swap the model; the memory stays.

"Why won't iFixit do this?"

iFixit knows the device. We know your bench, your parts, and your proof. The manual is static; the state is live.

"Is repair big enough?"

Repair is the wedge. Object permanence is the platform: assembly, maintenance, lab protocols, manufacturing, surgery.

"Can it work live?"

The bench is designed so the model can't fail; the graph and passport are real; the demo is rehearsed, not faked.

What exists, and what is actually new.

Repair tools are either static manuals or stateless chatbots. Perception models can segment a clip but do not maintain state. Cairn is the first system to close the loop between the manual, the object, and the proof.

EXISTING THING	WHAT IT DOES	WHAT IT LACKS
iFixit / repair manuals	Static guide for a known device.	Does not know **your** device's state, what you removed, or whether you reassembled correctly.
AR repair tools	Authored holograms and step overlays.	Not zero-shot; requires content production per device; no persistent object state.
GPT repair chatbots	Answer questions about a device.	Stateless; cannot see the bench, track parts, or verify the outcome.
SAM demos	Segment and track objects in a clip.	No object graph, no task reasoning, no verification, no repair passport.
Field-service proof tools	Capture evidence and closeout packets.	Not live perception; not object-level state; manual data entry.

The novel mechanism

A persistent, verifiable object-state graph + a closed verification loop. SAM gives perception; the graph gives memory; Wonder gives reasoning; the invariants give trust. No single model does this. The composition is the contribution.

Why not just iFixit + GPT?

- iFixit knows the device; it does not know **your** bench.
- GPT can answer; it cannot see or remember.
- Cairn closes the loop between the manual, the physical object, and the proof.

Defensibility: why this is a company, not a demo

01

The passport network

Every repair produces a signed record. Over time, Cairn becomes the trust layer for refurbished devices — a "Carfax for repairs."

02

The object graph data

Every teardown improves the part map: which screws belong where, which steps are fragile, which devices fail. The data compounds.

03

Local-first privacy

Shops will not upload customer devices to a cloud. Glimmer-on-device is a wedge that generic cloud tools cannot copy cheaply.

> "The category is object permanence. The wedge is repair. The moat is the proof."
>
> POSITIONING

The flywheel

Repair → **Passport** → **Trust** → **More shops** → **More data** → **Better repair**

The wedge

Repair. Independent shops. Laptops, appliances, and the devices people depend on.

The platform

Any physical task with parts and state: assembly, maintenance, lab protocols, manufacturing, surgery.

Three minutes that win the room.

The demo is not a feature tour. It is a story with three acts: a mistake, a memory, and a proof. Record it five times; pick the best.

TIME	VISUAL	VOICEOVER / ACTION
0:00	Close-up: a laptop, a tray of screws, a camera overhead.	"This is a laptop. AI can design one, but it can't take one apart. Here's why."
0:15	Split screen: a label appears on a screw, then disappears as the screw leaves frame.	"AI has object recognition. It doesn't have object permanence. It forgets a part the moment it leaves the frame."
0:30	Teardown begins. Overlay populates: part count, tray map, next step.	"Cairn watches the bench. Every part gets an identity, a location, and a state."
1:00	You drop a screw. The count flickers; Cairn flags it.	"That's a hinge screw. It's not in the tray. Cairn noticed before I did."
1:15	You reach for the SSD without disconnecting the battery. Cairn blocks the step.	"Stop. Disconnect the battery first. This is a safety invariant, not a suggestion."
1:30	You pick up the wrong screw. Cairn redirects to tray slot 4.	"That's a 5mm hinge screw. That hole takes a 3mm. Try tray slot 4."
1:45	You walk away. Cut to a clock. You return. The graph is still there.	"I left for lunch. Cairn didn't. 14 parts removed. Three screws unaccounted for. The battery is still connected."
2:05	Reassembly. Each step checks green. The tray empties.	"Reassemble. Every step verified. Every part accounted for."
2:20	Repair passport: a clean document with timestamps, part counts, evidence frames.	"23 of 23 fasteners. No extra parts. Boot test passed. Repair passport signed."
2:35	Architecture flashes: SAM 3.1 → graph → Wonder → verification.	"SAM tracks. Wonder reasons. Glimmer keeps it local. This is object permanence for physical work."
2:55	Logo: CAIRN. Tagline.	"The first AI that can take something apart and put it back together."

Hero moment 1

The return. You leave and come back. The agent reconstructs the entire state. This is the moment a chatbot cannot fake.

Hero moment 2

The wrong screw. The agent knows which hole it belongs to. This is object permanence made physical.

Hero moment 3

The passport. A signed record of the repair. This is the accountability layer that turns a demo into a product.

Production notes

- **Record five times.** Pick the take where the overlay is readable and the physical action is clean.
- **Show the screen and the hands in the same frame.** The magic is the connection between the two.
- **Caption every claim.** Judges may watch muted. Add large, simple captions for the three hero moments.
- **Have a backup.** A replay mode, a second device, and a pre-recorded clip in case the live demo fails.

Show

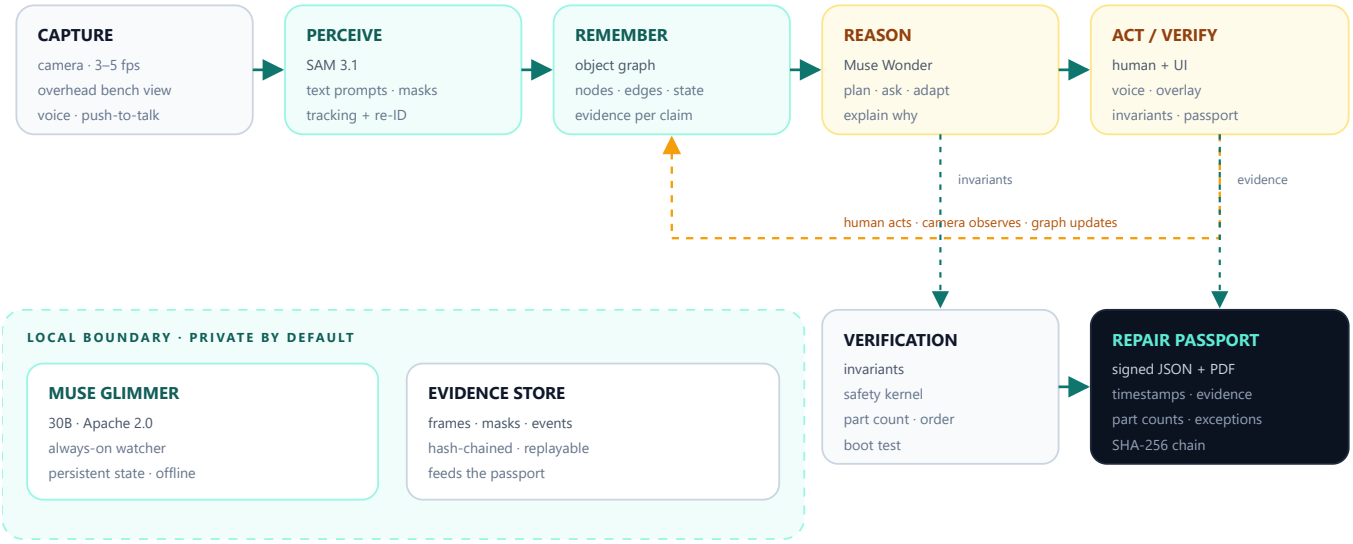
- The part count and the tray — the state is the story.
- The return after the break — the memory moment.
- The passport — the proof, on screen, at the end.

Don't show

- Prompt text, API calls, raw masks, or a terminal.
- Architecture before the payoff — earn the explanation.
- A live debug session. If it breaks, cut to the backup.

A loop, not a pipeline.

The camera is the input. The object graph is the product. The verification loop is the moat.



Deterministic skeleton

The task graph for the demo device is curated. The LLM chooses among valid steps; it does not invent destructive ones. This is how the demo survives.

LLM muscles

Muse Wonder handles language, explanation, adaptation, and interruption recovery. The graph handles truth. Neither replaces the other.

Human is the actuator

Cairn never touches the device. It proposes, observes, and verifies. The human acts. This is a copilot, not a robot — and that is a feature.

The safety kernel

Hard-coded rules the model cannot override: **disconnect power before removing components**, **never pry near a battery**, **ESD strap required**, **stop on smoke, heat, or swelling**. The LLM can explain a rule; it cannot disable it. This is the difference between a demo and a product.

One loop, 20 seconds

0.0 – 0.4 S
Perceive. Frame in, SAM masks out, graph diffed against the last known state.

0.4 – 1.2 S
Remember. New nodes, edges, and evidence frames are appended. Nothing is overwritten.

1.2 – 1.7 S
Reason. Wonder queries the graph, picks the next valid step, and explains why it matters.

1.7 S +
Act & verify. Voice and overlay respond; the human acts; invariants check the result. Repeat.

The loop is event-driven, not continuous: a change on the bench wakes it, and it sleeps again when the scene is stable.

Latency budget

STAGE	BUDGET
SAM track	< 400 ms / frame
Graph update	< 50 ms
Wonder next step	< 1.5 s
Voice round trip	< 500 ms
Verification	< 100 ms

Cached vs. live

Cached: task graph, part labels, SAM prompt templates, tray map, safety rules.

Live: masks, graph updates, next-step reasoning, voice, verification. The cache is what keeps the loop under two seconds.

Memory you can audit.

Every part is a node. Every relationship is an edge. Every claim has an evidence frame. The graph is the single source of truth for the agent, the UI, and the passport.

Node types

NODE	ATTRIBUTES
Part	id, class, label, size, state, location, confidence
Tool	id, type, size, location, in_use
Step	id, order, action, preconditions, safety
Observation	frame, mask, timestamp, model, confidence
Exception	type, severity, resolved, evidence

Edge types

EDGE	MEANING
fastens	Part is a fastener for another part.
belongs_to	Part is part of a subassembly.
connects	Cable or connector joins two parts.
requires	Step requires a tool or precondition.
precedes	Ordering constraint between steps.

```
// a part in the graph
{
  "part_id": "p_04",
  "class": "fastener",
  "label": "M2×3 hinge screw",
  "state": "removed",
  "location": "tray_04",
  "confidence": 0.97,
  "evidence": ["frame_01842.jpg"],
  "edges": [
    {"type": "fastens", "to": "hinge_bracket"},
    {"type": "requires", "to": "tool_ph0"}
  ]
}
```

Why a graph, not a table

- **Relationships are the task.** "Which screw fastens this bracket?" is a graph query, not a row lookup.
- **Evidence is first-class.** Every node points to the frame that justified it. The passport can cite it.
- **State is temporal.** Events are appended, never overwritten. You can replay the repair.
- **Exceptions are data.** A missing part is a node, not an error message.

The repair state machine

SEALED → OPENED → POWER_OFF → COMPONENTS_OUT → REASSEMBLING → VERIFIED ↘ EXCEPTION

Verification invariants

- **Count:** every removed part is either reinstalled or explicitly retired.
- **Order:** no step runs before its preconditions.
- **Safety:** power is off before any component removal.
- **Closure:** the tray is empty; no extra parts remain.
- **Function:** the device boots and passes the smoke test.

The passport

A signed JSON + PDF bundle: device ID, technician, timestamps, part counts, exception log, evidence frames, and a SHA-256 chain. It is the artifact that turns a repair into a verifiable record — and the seed of the business.

Three queries that run the product

QUERY	RETURNS	USED BY
missing_parts(device)	[p_04, p_11]	Exception banner
next_step(device)	"disconnect battery"	Voice + overlay
can_close(device)	false — 2 fasteners unaccounted	Passport gate

Every event is append-only, so the entire repair can be replayed from the passport.

Boring stack, exotic capability.

Use the simplest tools that let the capability shine. The novelty is in the memory and the loop, not in the framework.

LAYER	CHOICE	WHY
Capture	Phone or webcam; WebRTC; 3–5 fps sampler	Enough temporal resolution for tracking; low bandwidth; works on a laptop.
Perception	SAM 3.1 via Meta Model API	Zero-shot text prompts, pixel masks, identity-preserving video tracking.
Memory	SQLite + JSON graph (or Neo4j Aura free)	Persistent, queryable, easy to replay. No infrastructure to manage.
Reasoning	Muse Wonder (`muse-wonder-1.0`)	Long-horizon planning, tool use, active memory, error recovery.
Fast path	Muse Spark 1.3	Cheap classification, extraction, and short responses; saves credits.
Voice	Muse Voice Transcribe	Hands-free commands, turn detection, optional diarization for expert mode.
Visuals	Muse Image	Annotated step diagrams and passport illustrations; not on the critical path.
Local	Muse Glimmer 30B via Ollama / llama.cpp / vLLM	Always-on watcher; private by default; qualifies for the open-weights bonus.
UI	Next.js + a canvas overlay	Fast to build; readable on video; no native app needed.

Model pricing (list)

MODEL	PRICE
Muse Wonder	$4 input / $20 output per 1M tokens
SAM 3.1 — image	$2.50 / 1k images
SAM 3.1 — video	$0.20 / 1k frames
Muse Glimmer	Open weights; your hardware or partner

Credit math: the demo is cheap

A 10-minute teardown at 5 fps ≈ **3,000 frames**. At $0.20 / 1k frames, that is **$0.60** of SAM tracking. A 20-minute agent session with 200k output tokens is **~$4** of Muse Wonder. The $150 credit is enough for the build, the eval, and ten demo takes — if you cache and use Spark for the fast path.

Local deployment

Glimmer 30B runs on a single consumer GPU or a Mac. Quantize for a laptop; use Modal/Baseten for the demo if the laptop is thin.

Privacy story

The camera feed and object graph stay on the bench. Only high-level planning calls Muse Wonder, and that prompt contains no customer video.

Open-weights bonus

The always-on watcher is a real Glimmer workload, not a token integration. Document the endpoint, latency, and quantization.

What we are not building

A robot arm. A cloud video archive. A generic repair chatbot. A dashboard. The human stays the actuator; the graph stays on the bench; the passport is the only thing that leaves.

Ten days, one wedge.

The plan is ordered by risk, not by feature. Day 1 is a kill test. If perception fails, the idea changes before you write a UI.

DAY 1	**Kill test.** Can SAM 3.1 track 6–8 part classes across a real teardown with acceptable ID switches? If not, change the device, the lighting, or the tray. Do not proceed until this is yes.
DAY 2	**Object graph.** Schema, CRUD, event log. Ingest the first teardown. Prove you can answer "what is missing?" from the graph alone.
DAY 3	**Agent loop.** Muse Wonder + tool calls: query graph, request a SAM prompt, propose the next step. Structured outputs only.
DAY 4	**Voice + hands-free.** Push-to-talk, turn detection, interruption. Tune latency until the conversation feels natural.
DAY 5	**Verification.** Invariants, safety kernel, repair passport. The passport is a product artifact, not a debug log.
DAY 6	**UI polish.** Overlay, tray map, timeline, passport view. Make the state readable in a 1080p screen recording.
DAY 7	**Local Glimmer.** Deploy the always-on watcher; add a "privacy mode" toggle. Document the endpoint and latency.
DAY 8	**Eval + hardening.** Ten recorded sessions; measure recall, ID switches, missing-part latency. Fix the top three failure modes.
DAY 9	**Demo video.** Record five takes; pick the best; add captions; cut to 3:00. Do not leave this to the last day.
DAY 10	**Buffer + submit.** Re-read the rubric; check the checklist; submit early. Keep the backup footage.

What to cut

- Multi-device generalization.
- Full AR / glasses integration.
- A mobile app or a landing page.
- A business-model deck (put it in the appendix).
- Any feature that does not appear in the 3-minute video.

Team of four

Perception: SAM, tracking, calibration, tray design.
Agent/backend: graph, state machine, Wonder tool calls.
Frontend/UX: overlay, passport, video capture.
Domain/demo: repair task graph, safety rules, physical setup.

Solo? Cut to perception + graph + a simple overlay. Use Muse Code to scaffold the UI.

The one rule

If a feature does not make the demo more reliable, more visual, or more memorable, it is not in scope. Ten days is enough for one wedge — not for a platform.

The fallback if perception fails

Switch to human-confirmed memory: the user taps the part on the tray, the graph records it. The demo still works — it becomes "memory + verification," not "autonomous perception." The product survives; the story survives.

What survives: the object graph, the invariants, the passport, and the return-after-lunch moment. **What changes:** the input is a tap instead of a mask. Nothing the judges score depends on full autonomy.

Go / no-go gates

GATE	QUESTION	IF NO
Day 1	6–8 classes at ≥95% recall?	Change device, light, tray
Day 5	Three hero moments work?	Cut to two; polish passport
Day 8	Demo video is clear?	Simplify and re-record

If you only have 48 hours.

The full blueprint is 10 days. The winning demo is 48 hours. Build the three hero moments, record them, and tell the platform story in the pitch.

HOURS	BUILD	DEFINITION OF DONE
0–6	Pick the device; set up the bench; overhead camera; labeled tray.	A clean 1080p feed; 6–8 parts are visible and separable.
6–12	SAM 3.1 integration; prompt each part class; verify tracking.	Part count is stable across a 5-minute teardown.
12–20	SQLite graph; state machine; overlay with count + next step.	"What is missing?" works from the graph.
20–28	Muse Wonder loop; structured tool calls; three scripted interactions.	Drop, skip, wrong-screw all handled without a crash.
28–36	Verification + passport; simple PDF/JSON output.	A signed passport with timestamps and part counts.
36–42	Record five takes; pick the best; add captions; cut to 3:00.	A video a stranger understands without reading.
42–48	Submit; buffer; backup footage; final rubric check.	Submitted early, with a backup plan.

The minimum demo

- **One device:** a desktop PC or a simple laptop with 6–8 parts.
- **Three interactions:** missing screw, wrong order, reassembly check.
- **One artifact:** the repair passport.
- **One memory moment:** walk away and return.

Design the environment

- **Overhead camera, bright light.** Shadows kill tracking.
- **Labeled tray with wells.** The tray is the memory; the model just reads it.
- **Slower movements.** A demo is not a race; motion blur is the enemy.
- **Larger parts for live.** Use screws for the story, bigger parts for the tracking.

What to fake — and what not to

Fake the environment: stage the bench, label the tray, slow the motion, choose a device with clean separations. That is product design. **Do not fake the results:** the graph, the counts, the passport, and the memory must be real. Judges can smell a staged result; they respect a designed demo.

The three-frame storyboard

01

The mistake

A dropped screw, a skipped step, a wrong screw. The problem is visible in five seconds.

02

The memory

Walk away, return, the graph is still there. This is the moment a chatbot cannot fake.

03

The proof

The passport: counts, timestamps, boot test. The repair becomes a record.

Show the measurement, not just the magic.

Technical depth is 20% of the score. The fastest way to signal it is a small, honest eval table. Most teams will not have one.

METRIC	TARGET	HOW TO MEASURE
Part detection recall	≥ 95%	Ground-truth count per session vs. graph nodes created.
ID switch rate	≤ 5%	Manual review of 10 clips; count identity changes across occlusion.
Missing-part latency	≤ 5 s	Time from the part leaving frame to the exception node.
Wrong-order detection	100% on scripted errors	Run 10 known violations; count catches.
False alarm rate	≤ 10%	Clean sessions with no errors; count spurious flags.
Task completion time	≤ 1.3× baseline	Teardown with Cairn vs. without; do not slow the technician down.

Test protocol

1. Record 10 teardowns of the same device, 3 technicians, 2 lighting setups.
2. Log every part, every removal, every exception with timestamps.
3. Compute the metrics above; keep the raw clips for review.
4. Publish the table in the submission — including the failures.
5. Show one failure and how the system recovered. Honesty reads as craft.

The evidence table

RUN	PARTS	CAUGHT	MISSED	FALSE
01	23	23	0	1
02	23	22	1	0
03	23	23	0	0
…	…	…	…	…

Put this in the submission. It is more convincing than another feature bullet.

> "A demo shows what the system can do. An eval shows what it can be trusted to do."
>
> THE TECHNICAL-DEPTH ARGUMENT

Failure taxonomy

FAILURE	LOOKS LIKE	LOG / FIX
Perception	Part not detected; ID switch	Frame, mask, confidence; re-prompt
Memory	Missing part; duplicate node	Graph diff; reconcile
Reasoning	Wrong next step; skipped safety	Task-graph violation; hard stop
Action	Human does the wrong thing	Invariant check; voice correction

Rig

One overhead camera, a printed scale marker, a labeled tray, and constant light. No lab, no motion capture.

Ground truth

A paper checklist per session, marked by a second person. One page, twelve ticks, timed.

Reporting

A JSON metrics file committed to the repo, plus the table and one failure clip in the submission.

Assume it will break. Design for it.

The riskiest assumption is not the LLM. It is perception: can SAM 3.1 track the parts reliably enough for the story? Test that first.

HIGH IMPACT · LIKELY

1. Small-part tracking fails
2. Live demo crashes
3. Agent invents a step

HIGH IMPACT · UNLIKELY

4. Credits run out
5. Local GPU too slow

LOW IMPACT · LIKELY

6. UI feels busy
7. Video too long

LOW IMPACT · UNLIKELY

8. Judge asks about a different device

RISK	MITIGATION
Small-part tracking fails	Overhead camera, bright light, labeled tray, 6–8 classes, larger parts for live. Design the bench so the model can succeed.
Agent invents a step	Curated task graph; structured outputs; human confirmation before destructive steps; safety kernel the model cannot override.
Live demo crashes	Pre-record the 3-minute video; replay mode; second device; rehearsed chaos test; backup clip on a local drive.
Credits run out	SAM video is ~$0.60 per 10-minute teardown; use Spark for the fast path; cache prompts; Glimmer local for the watcher.
Local GPU too slow	Quantize; or deploy Glimmer on Modal/Baseten for the demo and keep the "local" claim honest by showing the endpoint.
"Just an image analyzer"	Lead with memory and verification. Show the return, the wrong screw, and the passport — not the segmentation mask.

Demo insurance checklist

- [ ] Three recorded takes, one chosen, all backed up.
- [ ] A replay mode that works without the camera.
- [ ] A second device and a second tray.
- [ ] A local copy of the passport and the eval table.
- [ ] A one-page script with the three hero moments.
- [ ] A 30-second clip for the viral bonus.

What could make us lose

IF…	THEN…
The demo is a chat window	We lose the room in the first 20 seconds.
The graph is not visible	We look like a SAM wrapper.
The passport is faked	We lose the only thing that makes us trustworthy.

The recovery rule

If the system and the technician disagree, the technician wins — and Cairn records the disagreement as an exception with its evidence. Trust is built by conceding in public, not by insisting.

A wedge with a platform story.

The hackathon is won on the demo, but the idea is stronger if the judges can see the company behind it. Start with the people who feel the pain every day.

01

Beachhead

Independent repair shops. 1–10 techs, high device mix, thin margins, no training budget. Sell per-tech seats.

02

Expansion

ITAD and refurbishers. Thousands of devices, warranty and resale-grade proof. Sell per-device certification.

03

Platform

OEM warranty and insurance. A verified repair record reduces fraud and disputes. License the passport standard.

Pricing hypothesis

SEGMENT	MODEL
Consumer	Free / $9.99 per month
Repair shop	$49–$99 per tech / month
ITAD / refurbisher	$2 per certified device
OEM / insurer	Annual license + audit API

The moat

- **The passport network.** Every repair becomes a trust record. A "Carfax for repairs" is hard to copy once the network exists.
- **The object graph data.** Every teardown improves the part map, the failure modes, and the task graph.
- **Local-first privacy.** Shops will not upload customer devices. On-device Glimmer is a structural advantage.

12-month roadmap

Repair copilot MVP → Shop SaaS + passport → ITAD certification → OEM audit API → Assembly / maintenance platform

> "Start with the bench. End with the standard for how physical work is remembered."
>
> THE LONG GAME

Validate while you build

01

5 shops

Call five independent repair shops. Ask one question: "What do you lose track of during a teardown?"

02

20 devices

Run Cairn on 20 real devices. Measure time-to-repair and lost-part rate against the baseline.

03

3 metrics

Lost parts, repeat repairs, tech ramp time. If these move, the business is real.

Unit economics, distribution, and the ask

Unit economics

A seat is $79 / tech / month. One avoided repeat repair — a part, a board, or an hour — pays for it. Gross margin stays high because the expensive work runs on the shop's own GPU.

Distribution

Repair forums, iFixit communities, trade schools, and ITAD associations. The 30-second clip is the top of the funnel; the passport is the reason to stay.

The ask

Next 90 days: 20 shops, 500 certified devices, one ITAD pilot. That is enough to prove retention and price before raising.

Say it in one breath.

The judges will read the submission before they watch the video. Make the first paragraph do the work.

The submission paragraph

"Cairn gives AI object permanence for physical repair. Today's models recognize a screw but forget it the moment it leaves the frame, so they cannot guide a real teardown. Cairn uses SAM 3.1 to detect and track every part with identity, Muse Wonder to plan and reason over a persistent object graph, and Muse Glimmer to run locally so a shop's video never leaves the bench. In the demo, a technician disassembles a laptop, drops a screw, skips the battery step, and walks away — Cairn catches all three and certifies the reassembly with a signed repair passport. It targets the 62M-tonne e-waste problem and the repair skills gap."

The 30-second video script

"AI can design a phone. It can't take one apart. Why? Because it has object recognition — but not object permanence. It forgets a part the moment it leaves the frame. Cairn fixes that. It watches the bench, tracks every part, remembers the teardown, and proves the reassembly. Drop a screw. Skip the battery. Walk away. Cairn remembers. This is object permanence for physical work."

The X hook

"AI can design a laptop but can't take one apart. We fixed the missing sense: object permanence."

Thread outline: 1) the amnesia problem, 2) the demo clip, 3) how SAM + Wonder + Glimmer work, 4) the repair passport, 5) the e-waste number, 6) the ask.

Submission checklist

- [ ] 3-minute demo video, captioned, with the three hero moments.
- [ ] One-paragraph pitch that names the problem, the capability, and the beneficiary.
- [ ] Architecture diagram and a short "how it works" section.
- [ ] Eval table with 10 runs, including one honest failure.
- [ ] Repair passport sample (JSON + PDF).
- [ ] Open-weights deployment note for the Glimmer bonus.
- [ ] 30-second viral clip and a thread draft.
- [ ] Backup footage, backup device, backup passport.

The closing line

"Don't compete on intelligence. Compete on state." Everyone can call a model. Almost no one builds a system that remembers, verifies, and acts over time in the physical world. That is the moat, the demo, and the story.

The same pitch at three lengths

10 seconds

"Cairn gives AI object permanence for repair — it remembers every part and proves the reassembly."

30 seconds

"AI can recognize a screw but forgets it when it leaves the frame. Cairn tracks every part, remembers the teardown, and catches the missing screw."

60 seconds

"AI can design a phone but can't repair one, because it has no memory of the physical world. Cairn gives it object permanence: SAM tracks, Wonder reasons, Glimmer stays local. The result is a signed repair passport."

TITLE	TRACK	BONUS POOLS	TAGS
Cairn — Object permanence for physical work	Indie Developer · Build for Your World	Viral · Open Weights · Community · Agentic · Creative	#repair #objectpermanence #SAM3.1 #local-first

The receipts.

Model capability matrix

MODEL	CAPABILITY USED	WHY IT IS NECESSARY
Muse Wonder	Long-horizon planning, tool use, active memory, error recovery.	Runs the repair loop; explains steps; adapts to interruptions.
SAM 3.1	Text-prompted detection, segmentation, identity-preserving video tracking.	The perception layer; zero-shot means no training data per device.
Muse Glimmer	30B open weights; local, always-on agents; persistent state.	Privacy, offline operation, and the open-weights bonus.
Muse Voice Transcribe	Speech-to-text, speaker diarization, turn detection.	Hands-free interaction; expert-mode narration.
Muse Image	Agentic image generation with planning and self-correction.	Passport illustrations and step diagrams.

Glossary

Object permanence: the understanding that objects continue to exist when they are out of sight. Here: a persistent, queryable model of every part across time.

Re-ID: re-identification of an object after occlusion or leaving the frame, preserving its identity.

Invariant: a property that must hold before the repair is considered complete.

Repair passport: a signed record of the repair: parts, steps, exceptions, evidence, and outcome.

Sources

Hackathon: dev.meta.ai/events/global-hackathon; cerebralvalley.ai/e/meta-ai-developer-hackathon (rubric, tracks, dates, prizes).

Models: dev.meta.ai/models/muse-wonder; /sam-3-1; /muse-glimmer; /muse-voice-transcribe; /muse-image; /muse-spark.

E-waste: Global E-waste Monitor 2024 (UNITAR) — 62M tonnes, 22.3% formally recycled.

Skills gap: Deloitte / The Manufacturing Institute — 1.9M projected unfilled US manufacturing jobs by 2033.

Verify all figures against the primary source before putting them on a slide. Dates and prizes may change at Meta's discretion.

> "The model is not the product. The memory is."
>
> CAIRN · BLUEPRINT V1.0

Working name

Cairn. Alternatives: Keel, Latch, Tenon. The name matters less than the sentence: "The first AI that can take something apart and put it back together."

The one-liner

Cairn gives AI object permanence for physical work — starting with repair.

The bet

The next wave of AI is not more intelligence. It is memory, verification, and agency in the physical world. Cairn is a bet on that wave.

Pre-submission verification

- [ ] Demo video is under 3:00, captioned, and playable without sound.
- [ ] The three hero moments are each readable in under 10 seconds.
- [ ] Part counts, timestamps, and passport hashes match the recording.
- [ ] Rubric line is answered explicitly: Impact, Tooling, Depth, Demo.
- [ ] Every claimed model has a visible, necessary role in the video.
- [ ] All figures cite a primary source; no stat is older than its claim.

DOCUMENT	VERSION	TRACK	LENGTH
Cairn — Project Blueprint	1.0 · 2 November 2026	Indie Developer	18 sections · A4